

MINI PROJECT SYNOPSIS

1. Title of the Project:

Smart Priority-Based Wireless Traffic Control System

2. Introduction:

In modern urban environments, traffic management requires a balance between automated efficiency and manual intervention for emergency or high-density situations. Traditional systems often lack the flexibility to switch between a fixed timer and a priority-based override. This project develops a dual-mode system using a wireless transmitter (Arduino) to send directional priority signals to a receiver-side traffic light array, ensuring smooth transitions and safety buffers.

3. Problem Statement:

Many existing traffic systems are either fully manual or strictly automated. The current challenge in this prototype is the failure of the "Normal Cycle" to execute autonomously without user input. A reliable system must maintain a continuous real-time cycle while remaining "interrupt-ready" to process wireless priority requests without crashing the sequence or causing abrupt signal changes.

4. Objectives:

- To implement a Normal Automated Cycle (Green-Yellow-Red sequence) for four directions.
- To establish a wireless link where a Red LED on the transmitter confirms signal dispatch and a Blue LED on the receiver confirms signal reception.
- To create a Priority Override that finishes the current active timing before switching to the requested direction.
- To ensure an "All-Red" Safety Buffer (2-3 seconds) between transitions to prevent accidents.
- To automatically resume the normal cycle once the priority duration ends.

5. Proposed Methodology:

The system will use a State Machine logic instead of linear delays. This allows the Arduino to "listen" for button presses while simultaneously counting time for the LEDs.

Operational Logic:

1. Default Mode: The system runs a loop (Directions 1→2→3→4). Each transition follows:
Green →Yellow →All-Red→Next Green
2. Interrupt Mode: When a button is pressed on the Transmitter:
 - The Transmitter Red LED flashes.
 - The Receiver Blue LED turns ON.
 - The system checks the current state. It completes the minimum timing for the current Green, moves to Yellow, then switches to the requested direction's Green.
3. Return Path: Once the priority timer expires and no other buttons are pressed, the system remembers its last position in the normal cycle and resumes from there.

6. Hardware Requirements

1. Microcontrollers: 2x Arduino (One for Transmitter, one for Receiver).
2. Input: 4x Push Buttons (for North, South, East, West directions).
3. Transmitter Indicators: 1x Red LED (Signal Sent).
4. Receiver Indicators: * 12x LEDs (4 Directions $\times$ Red, Yellow, Green lights).
5. Communication: RF Module (433MHz) or NRF24L01 for wireless data transfer.
6. Power: 5V DC Supply.

7. Software Requirements:

1. Arduino IDE for coding.
2. Embedded C using `millis()` for non-blocking timing.
3. Logic: Conditional State Switching.

8. Expected Outcomes:

- Real-time monitoring of environmental parameters.
- Wireless data transmission
- Alert system for abnormal conditions.
- Low-cost and scalable design suitable for various applications.

9. Applications:

- Smart Cities
- Emergency Handling
- Accident free Roads

10. Future Scope:

- Integration with AI for predictive analysis.
- Solar-powered operation for remote areas.
- Development of a mobile application with advanced analytics.

11. References:

1. Research papers on IoT-based Traffic Control Systems.
2. ESP32 technical datasheet.